

SQL Query Analysis Results

Module 3 Assignment

--- Question 1 ---

Question: How many entries in your database are from applicants who applied for Fall 2026?

Result: Fall 2026 applicant count: 33,207

SQL Query:

```
SELECT COUNT(*) FROM applicants WHERE term ILIKE '%Fall 2026%';
```

Explanation:

This query uses the COUNT(*) aggregate function to calculate the total number of rows. The WHERE clause filters the dataset to only include records where the term column contains the string "Fall 2026", utilizing ILIKE for case-insensitive matching.

--- Question 2 ---

Question: Among entries that provide a nationality classification, what percentage are international students?

Result: Percent international: 47.89%

SQL Query:

```
SELECT
    (SUM(CASE WHEN us_or_international ILIKE 'International' THEN 1 ELSE 0 END) * 100.0) /
    NULLIF(COUNT(us_or_international), 0)
FROM applicants
WHERE us_or_international IS NOT NULL
    AND TRIM(us_or_international) != '';
```

Explanation:

This calculates a percentage entirely within SQL. The denominator uses COUNT to find all valid nationality entries. The numerator utilizes a CASE statement to assign a 1 to 'International' applicants and 0 to others, summing them together.

--- Question 3 ---

Question: What are the average GPA, GRE Quantitative, GRE Verbal, and GRE Analytical Writing scores of applicants who provide each metric?

Result: Average GPA: 3.77 | Avg GRE Quant: 261.45 | Avg GRE Verbal: 161.14 | Avg GRE AW: 9.17

SQL Query:

```
SELECT AVG(gpa), AVG(gre), AVG(gre_v), AVG(gre_aw) FROM applicants;
```

Explanation:

This uses SQL's built-in AVG() aggregate function. Because AVG() automatically ignores NULL values by design, it perfectly satisfies the requirement to calculate averages using only the applicants who provided that specific score.

--- Question 4 ---

Question: What is the average GPA of American applicants who applied for Fall 2026?

Result: Average GPA of American applicants (Fall 2026): 3.79

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SQL Query:

```
SELECT AVG(gpa) FROM applicants  
WHERE term ILIKE '%Fall 2026%' AND us_or_international ILIKE 'American';
```

Explanation:

The query calculates the average of the gpa column while applying two strict conditions in the WHERE clause using ILIKE for case-insensitive matching.

--- Question 5 ---

Question: What percentage of Fall 2025 entries are acceptances?

Result: Fall 2025 acceptance percentage: 40.74%

SQL Query:

```
SELECT (SUM(CASE WHEN status ILIKE '%Accept%' THEN 1 ELSE 0 END) * 100.0) / NULLIF(COUNT(*), 0)  
FROM applicants WHERE term ILIKE '%Fall 2025%';
```

Explanation:

The WHERE clause isolates Fall 2025 applications. The CASE statement flags any status containing "Accept". Dividing this sum by the total count yields the percentage.

--- Question 6 ---

Question: What is the average GPA of accepted applicants who applied for Fall 2026?

Result: Average GPA of accepted applicants (Fall 2026): 3.78

SQL Query:

```
SELECT AVG(gpa) FROM applicants  
WHERE term ILIKE '%Fall 2026%' AND status ILIKE '%Accept%';
```

Explanation:

This takes the average of the gpa column and filters for Fall 2026 applicants whose status field contains the word "Accept".

--- Question 7 ---

Question: How many entries are from applicants who applied to Johns Hopkins University for a master's degree in Computer Science?

Result: JHU Master's CS applicant count: 18

SQL Query:

```
SELECT COUNT(*) FROM applicants  
WHERE (program ILIKE '%Johns Hopkins%' OR program ~* '\bJHU\b')  
AND program ILIKE '%Computer Science%' AND degree ILIKE '%Master%';
```

Explanation:

This filters text using multiple ILIKE statements and a PostgreSQL case-insensitive regular expression (~* '\bJHU\b') with word boundaries to correctly capture the acronym "JHU".

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--- Question 8 ---

Question: How many Fall 2026 entries are acceptances from applicants applying for a PhD in Computer Science at Georgetown, MIT, Stanford, or Carnegie Mellon (using original fields)?

Result: Original-field count: 30

SQL Query:

```
SELECT COUNT(*) FROM applicants
WHERE term ILIKE '%Fall 2026%' AND status ILIKE '%Accept%'
  AND (degree ILIKE '%PhD%' OR degree ~* '\bPh\.?D\b') AND program ILIKE '%Computer Science%'
  AND (program ILIKE '%Georgetown%' OR program ILIKE '%Massachusetts Institute of Technology%' OR program ~*
'\bMIT\b' OR program ILIKE '%Stanford%' OR program ILIKE '%Carnegie Mellon%' OR program ~* '\bCMU\b');
```

Explanation:

This applies AND conditions to filter for Term, Status, Degree, and Major, and a grouped set of OR clauses to capture the specific university names or acronyms.

--- Question 9 ---

Question: Repeat Question 8 using the LLM-generated university and program fields, and compare the results.

Result: Original count: 30 | LLM count: 30 | Difference: 0

SQL Query:

```
SELECT COUNT(*) FROM applicants
WHERE term ILIKE '%Fall 2026%' AND status ILIKE '%Accept%' AND (degree ILIKE '%PhD%' OR degree ~*
'\bPh\.?D\b') AND llm_generated_program ILIKE '%Computer Science%'
  AND (llm_generated_university ILIKE '%Georgetown%' OR llm_generated_university ILIKE '%Massachusetts Institute
of Technology%' OR llm_generated_university ~* '\bMIT\b' OR llm_generated_university ILIKE '%Stanford%' OR
llm_generated_university ILIKE '%Carnegie Mellon%' OR llm_generated_university ~* '\bCMU\b');
```

Explanation:

This functions identically to Question 8 but targets the LLM columns. The difference of 0 indicates that for this narrow dataset, the original text was standard enough (or our regex robust enough) that the LLM data standardization did not yield hidden records.

--- Custom Question 1 ---

Question: What is the average GPA of Fall 2026 PhD applicants, grouped by their admission status?

Result: Wait listed: 3.84 (1,063) | Accepted: 3.83 (3,565) | Rejected: 3.80 (7,358)

SQL Query:

```
SELECT status, AVG(gpa) AS avg_gpa, COUNT(*) AS total_applicants
FROM applicants WHERE term ILIKE '%Fall 2026%' AND (degree ILIKE '%PhD%' OR degree ~* '\bPh\.?D\b') AND gpa
IS NOT NULL AND (status ILIKE '%Accept%' OR status ILIKE '%Reject%' OR status ILIKE '%Wait%')
GROUP BY status ORDER BY avg_gpa DESC;
```

Explanation:

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This uses GROUP BY to categorize applicants. It aggregates the average GPA and total volume per category, ordered from highest GPA to lowest.

--- Custom Question 2 ---

Question: Which 5 universities received the most Fall 2026 applications, and what is their average GRE Quant score?

Result: 1. Stanford (770, Avg GRE: 248.92) | 2. UC Berkeley (722, Avg GRE: 276.05) | 3. Yale (633, Avg GRE: 244.91) | 4. UWashingon (623, Avg GRE: 259.34) | 5. UW-Madison (604, Avg GRE: 237.85)

SQL Query:

```
SELECT llm_generated_university, COUNT(*) AS application_count, AVG(gre) AS avg_gre_quant
FROM applicants WHERE term ILIKE '%Fall 2026%' AND llm_generated_university IS NOT NULL
GROUP BY llm_generated_university ORDER BY application_count DESC LIMIT 5;
```

Explanation:

This groups applications by standardized LLM university names, calculates volume and average GRE score, orders by volume descending, and uses LIMIT 5 to isolate the top schools.